

Question- 1

Clone <https://github.com/zembutsu/docker-sample-nginx.git> on your local system.

- Initialize one repo on your local system named Cloudethix-nginx-yourname and then copy all the content from cloned repo to your local repo.
- Update the index.html and add the name CloudEthiX along with Hostname. Commit the change to your local repository. Then create and push the code from the local repository to your remote repository named Cloudethix-nginx-yourname under master branch.
- Once code is pushed, create a Docker hub public repository named Cloudethix-nginx-yourname.
- Then build the docker image out of the dockerfile. Once an image is built, tag the image and push that image to the docker hub repo you created.
- Create a deployment.yaml file with 2 replications in cloudethix_yourname Namespace and use the container image that you have pushed to your docker hub repo. Once a file is created, run the deployment in k8s cluster.
- Please prepare a document along with the snap and share it across

System Output

```
payal_kharat@LAPTOP # git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
payal_kharat@LAPTOP # ls
docker-sample-nginx
-----
payal_kharat@LAPTOP # mkdir Cloudethix-Nginx-Payal
payal_kharat@LAPTOP # cd Cloudethix-Nginx-Payal/
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-1/Cloudethix-Nginx-
Payal/.git/
-----
```

```
payal_kharat@LAPTOP # cp -rf docker-sample-nginx/* Cloudethix-Nginx-Payal/
payal_kharat@LAPTOP-# ls
Dockerfile  LICENSE  README.md  default.conf  index.html
```

```
payal_kharat@LAPTOP # git status
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
    Dockerfile
    LICENSE
    README.md
    default.conf
    index.html
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
payal_kharat@LAPTOP-# git add .
payal_kharat@LAPTOP-# git commit -m "Update index.html with CloudEthiX"
[master (root-commit) 9e6a3cb] Update index.html with CloudEthiX
 5 files changed, 48 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 LICENSE
 create mode 100644 README.md
 create mode 100644 default.conf
 create mode 100644 index.html
```

```
payal_kharat@LAPTOP-# git remote add origin
https://github.com/CloudEthix987/Cloudethix-Nginx-Payal.git
payal_kharat@LAPTOP-# git branch -M master
payal_kharat@LAPTOP-# git push -u origin master
Username for 'https://github.com': CloudEthix987
Password for 'https://CloudEthix987@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.40 KiB | 30.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/CloudEthix987/Cloudethix-Nginx-Payal.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

```
payal_kharat@LAPTOP-# podman build -t cloudethix-nginx:latest .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> bfba633494f5
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix-nginx:latest
--> 3634adf0fbe2
Successfully tagged localhost/cloudethix-nginx:latest
3634adf0fbe2aa22b60f8efae50b3ad6cd6a93e9639486c233d4b86c869e0377
```

```
payal_kharat@LAPTOP-# podman tag localhost/cloudethix-nginx
docker.io/payalkharat/cloudethix-nginx-payal:v1
payal_kharat@LAPTOP-# podman login docker.io
Username: payalkharat
Password:
Login Succeeded!
```

```
-----
payal_kharat@LAPTOP-# podman push docker.io/payalkharat/cloudethix-nginx-
payal:v1
Getting image source signatures
Copying blob 8cdfe4f23778 skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 0d62d88506ba skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob d0b4c8e075fc done    |
Copying blob da4dea1b00af skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying blob b83fc70afc5e skipped: already exists
Copying config 3634adf0fb done    |
Writing manifest to image destination
-----
```

```
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudethix-nginx
  namespace: cloudethix-payal
spec:
  replicas: 2
  selector:
    matchLabels:
      app: cloudethix-nginx
  template:
    metadata:
      labels:
        app: cloudethix-nginx
    spec:
      containers:
        - name: nginx
          image: docker.io/payalkharat/cloudethix-nginx-payal:v1
          ports:
            - containerPort: 80
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/cloudethix-nginx created
payal_kharat@LAPTOP-# kubectl get pods -n cloudethix-payal
NAME                                READY   STATUS    RESTARTS   AGE
cloudethix-nginx-f8db88b48-7gfwx   1/1     Running   0           17s
cloudethix-nginx-f8db88b48-j6t1p   1/1     Running   0           17s
payal_kharat@LAPTOP-# kubectl get deployment -n cloudethix-payal
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
cloudethix-nginx  2/2     2             2           79s
```